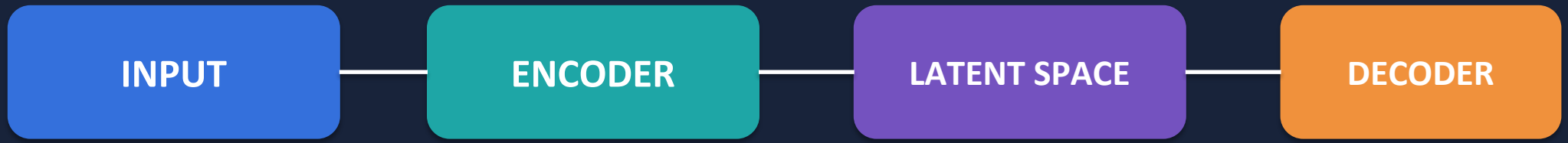
AUTOENCODERS

Learning Compact Representations from Data



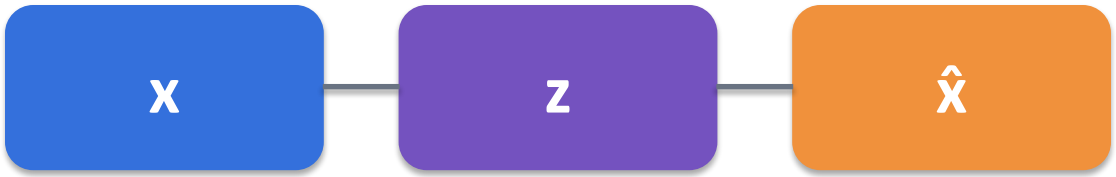
Compress → Represent → Reconstruct

Deep Learning • Unsupervised / Self-Supervised Representation Learning

1. What is an Autoencoder?

A neural network trained to reproduce its input

- Takes an input x and creates a compact representation z.
- The decoder uses z to reconstruct the input as $\hat{x}$.
- Training minimizes the difference between x and $\hat{x}$.
- It learns useful features without requiring class labels.



Encoder

Latent

Decoder

Loss = Difference(x, $\hat{x}$)

Key idea: learn what information is important enough to preserve.

2. Basic Architecture

The bottleneck forces the network to learn a representation



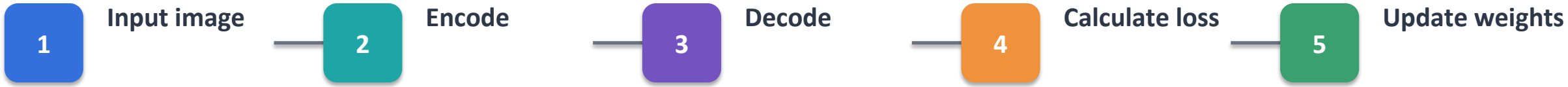
BOTTLENECK

A smaller latent space prevents simple copying and encourages feature extraction.

Example: 28×28 image = 784 values → 32 latent features → 784 reconstructed values

3. How Training Works

Forward pass → reconstruction → loss → backpropagation

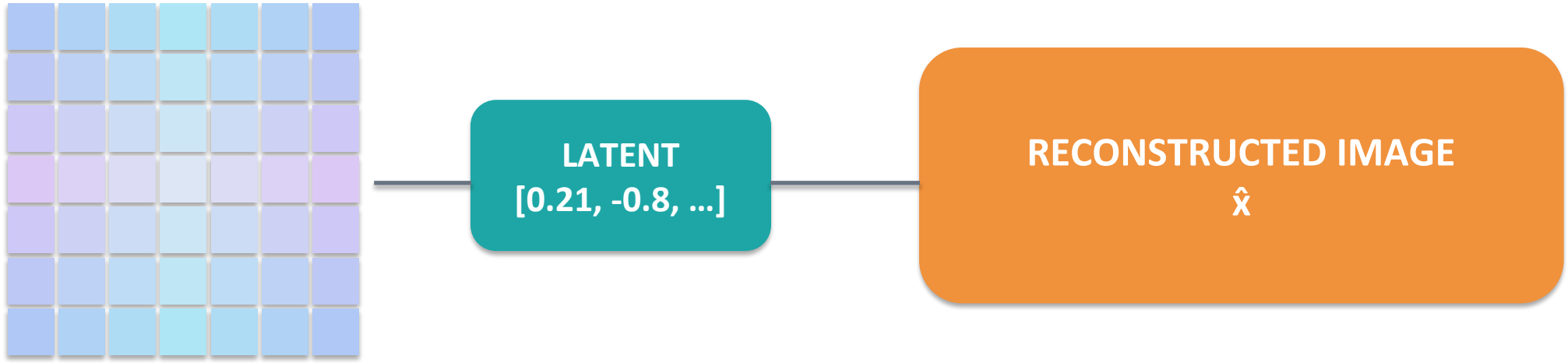


Loss ↓ as reconstructed output becomes closer to the original input

Common losses: Mean Squared Error (MSE) • Binary Cross-Entropy (BCE)

4. Intuitive Example: Compressing an Image

Think of the latent vector as a compact summary



The latent vector does not need to store every pixel. It captures patterns such as edges, shapes, textures and other useful structure.

5. Why Do We Need Autoencoders?

Common applications across computer vision and data analytics

1

Denoising

Remove noise from corrupted images

2

Anomaly Detection

Learn normal patterns; flag unusual samples

3

Dimensionality Reduction

Compress high-dimensional data

4

Feature Learning

Create useful representations for downstream ML

5

Compression

Store/transmit data using fewer latent values

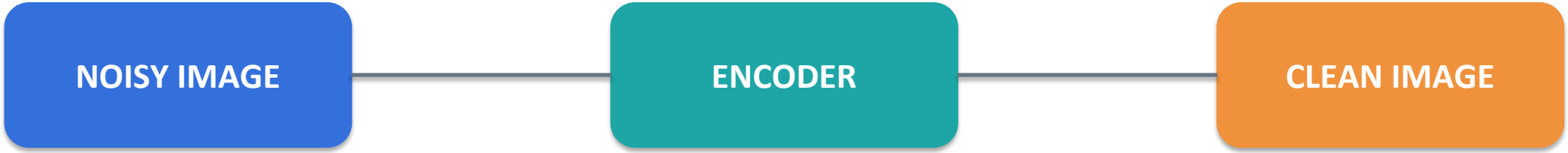
6

Generative Models

Foundation for Variational Autoencoders

6. Denoising Autoencoder

Input is deliberately corrupted; target is the clean version

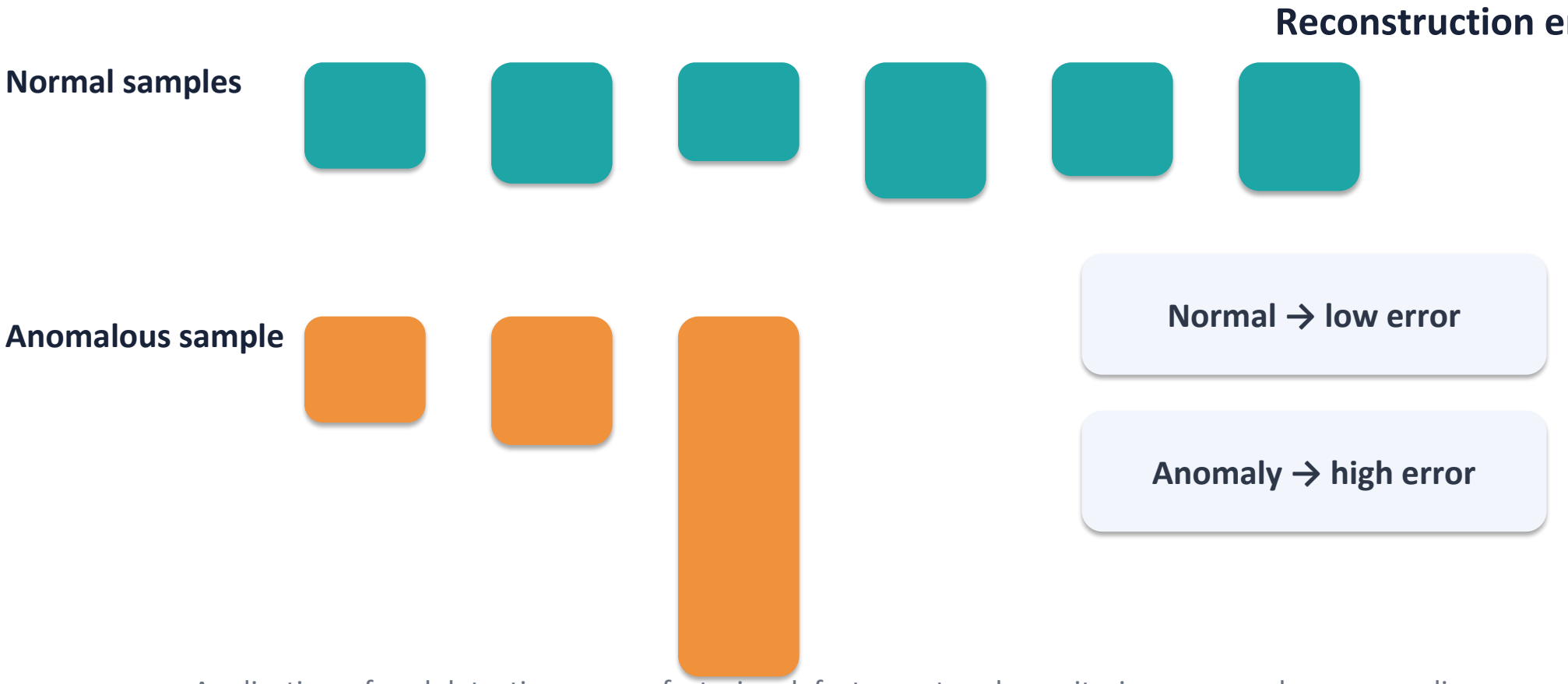


Training teaches the model to ignore irrelevant noise and preserve meaningful structure.

Example: noisy handwritten digit → clean handwritten digit

7. Autoencoder for Anomaly Detection

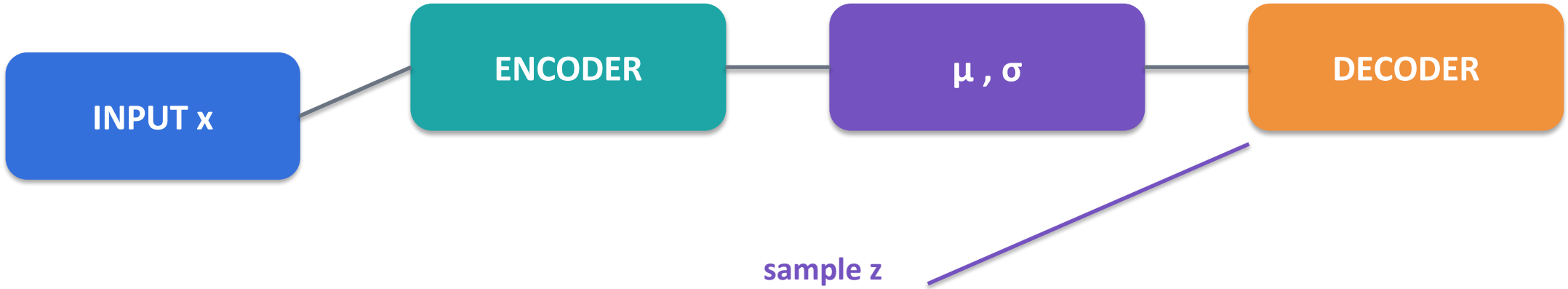
Reconstruction error can act as an anomaly signal



Applications: fraud detection • manufacturing defects • network monitoring • unusual sensor readings

8. Variational Autoencoder (VAE)

A probabilistic extension that learns a structured latent space



VAE learns a distribution rather than a single fixed code, making it possible to sample new data.

Loss = Reconstruction Loss + KL Divergence

9. Autoencoder vs PCA

Both can reduce dimensions, but they learn differently

Aspect	PCA	Autoencoder
Model	Linear transformation	Neural network
Relationship	Linear	Can be nonlinear
Features	Principal components	Learned latent features
Training	Variance reconstruction	Reconstruction loss
Scalability	Efficient for many classical tasks	Benefits from GPU/deep learning

Use PCA as a strong classical baseline; use autoencoders when nonlinear/deep representations are useful.

10. Simple Keras Example

A compact autoencoder for MNIST-style 28×28 images

```
encoder = keras.Sequential([
        layers.Flatten(),
    layers.Dense(128, activation="relu"),
layers.Dense(32, activation="relu")   # latent code
                    ])

decoder = keras.Sequential([
    layers.Dense(128, activation="relu"),
   layers.Dense(784, activation="sigmoid")
                    ])

autoencoder = keras.Sequential([encoder, decoder])
autoencoder.compile(optimizer="adam", loss="mse")
  autoencoder.fit(x_train, x_train, epochs=10)
```

Notice: the target is the input itself → x_train → x_train

TAKEAWAY

1. **Autoencoder = Encoder + Latent Space + Decoder**
2. It learns to reconstruct the input.
3. The bottleneck encourages compact feature learning.
4. Reconstruction error enables useful anomaly detection.
5. VAEs extend the idea toward probabilistic generative modeling.

Questions / Discussion